

Question: Compare RIP and OSPF as dynamic routing protocol options for this network, and recommend which one is better suited as the network grows, justifying your choice based on convergence time and scalability.

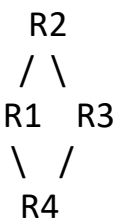
For a network with **five branch routers**, both RIP and OSPF can provide dynamic routing, but their behavior and scalability are different.

Feature	RIP	OSPF
Routing type	Distance-vector	Link-state
Metric	Hop count	Cost (based mainly on bandwidth)
Maximum hops	15 hops	No 15-hop limitation
Convergence	Relatively slow	Fast
Scalability	Suitable for small networks	Suitable for medium/large networks
Updates	Periodic routing updates	Sends updates mainly when topology changes
Network efficiency	More overhead as network grows	More efficient for larger networks
Complexity	Easier to configure	More complex

Convergence time

Convergence means how quickly all routers learn about a network change or failure.

Example:



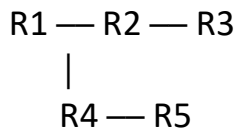
Suppose the link between **R1 and R2 fails**.

- **RIP:** Routers exchange routing information periodically, so it can take longer to recognize and adapt to the failure.
- **OSPF:** Routers detect the topology change and distribute link-state information, allowing the routing tables to be recalculated much faster.

So, **OSPF generally converges faster than RIP.**

Scalability

With five routers, RIP can still work:



But as the start up adds more branches, RIP becomes less suitable because of its **15-hop limit** and less efficient routing behavior.

OSPF is designed to scale much better. It can organize larger networks using **areas**, reducing the amount of routing information that each router needs to process.

Recommendation: OSPF

For this start up, I would choose **OSPF** as the network grows.

Main reasons:

1. ⚡ **Faster convergence** — It responds more quickly to link/network failures.
2. 📊 **Better scalability** — It can handle growth much better than RIP.
3. 🗺️ **Better path selection** — OSPF uses cost rather than simply counting hops.
4. 🔄 **Efficient updates** — It doesn't rely on constantly sending the complete routing table like traditional RIP operation.
5. 🏢 **Suitable for future expansion** — OSPF's area structure makes it appropriate as the number of branches increases.

RIP is simpler and can work well for small networks, but it has a 15-hop limitation and slower convergence. OSPF uses a link-state approach, provides faster convergence, and scales much better as the network grows. Since the start up is expanding from two routers to five and may continue growing, OSPF is the better choice because it provides faster recovery from topology changes and better scalability.